Absolute and Relative Age of Basalt Rock at Giant's Causeway

The absolute age of the hexagonal basalt rocks at Giant's Causeway in Ireland would be just under 60 million years ago, when Europe ripped away from North America and formed cracks that allowed molten rock and lava to come up. This lava and molten rock then cooled to form the basalt columns that are seen today at Giant's Causeway (*National Trust*, 2025). The relative age of the basalt rocks present at Giant's Causeway within the geologic time scale would be part of the Paleogene period, between the Eocene and Paleocene epoch (*Earth Science Conservation Review,* 2025). This would place the relative age to be between 56 million and 66 million years ago from present day. This is consistent with the absolute age, as within this period, an event known as the *Paleocene-Eocene Thermal Maximum* occurred, where much volcanic activity was happening across the world and raised the temperature of the Earth to its maximum for 100 000 years (*Encyclopedia Britannica,* 2025).

Figure 1

Giant's Causeway, courtesy of IUGS

Figure 1- Giant's Causeway basalt columns were formed when immense volcanic activity was occurring in Earth over 56 million years ago

How Basalt's Rock Age at Giant's Causeway was Determined

The basalt rock at Giant's Causeway had its age determined through the utilization of radiometric dating, using the argon method of radiometric dating (*Oxford Academic*, 2010). The argon method of radiometric dating is based off the potassium-argon dating, where a sample of the rock is exposed to radiation in a nuclear reactor with neutrons to convert the potassium isotope to an argon isotope. In the argon method, however, instead of potassium being the isotope, another isotope of argon is substituted in its place, and the age is calculated by taking the ratio of the argon isotopes. This is more efficient as this can be determined in a single experiment, instead of having to do two separate measurements (*Science Direct*, 2025). From this radiometric dating, researchers found that for the lower basalt formation, the age was about 62.6 million years old. For the upper basalt formation, the age was found to be about 59.6 million years old (*Oxford Academic, 2010*). From this data alone, it certifies the geological age being between the Eocene and Paleocene epoch. Another way that researchers have determined the age of Giant's Causeway was the fact that it was a part of the North Atlantic Igneous Province. This area, during the Paleocene era, was home to much volcanic activity and supports the claim that the basalt rocks were formed during this epoch, as the magma/lava cooled to form the basalt columns. Additionally, it was found that the basalt columns sit on top of older sediment (chalk), belonging to Upper Cretaceous epoch (*Hull Geological Society,* 2012). By the law of superposition, the basalt columns must be younger than the sediment that it is sitting on, and thus adds to the proof that the formation of this rock at Giant's Causeway was more recent relative to other rock formations in the area.

Figure 2

Map of North Atlantic Igneous Province (NAIP), courtesy of Research Gate

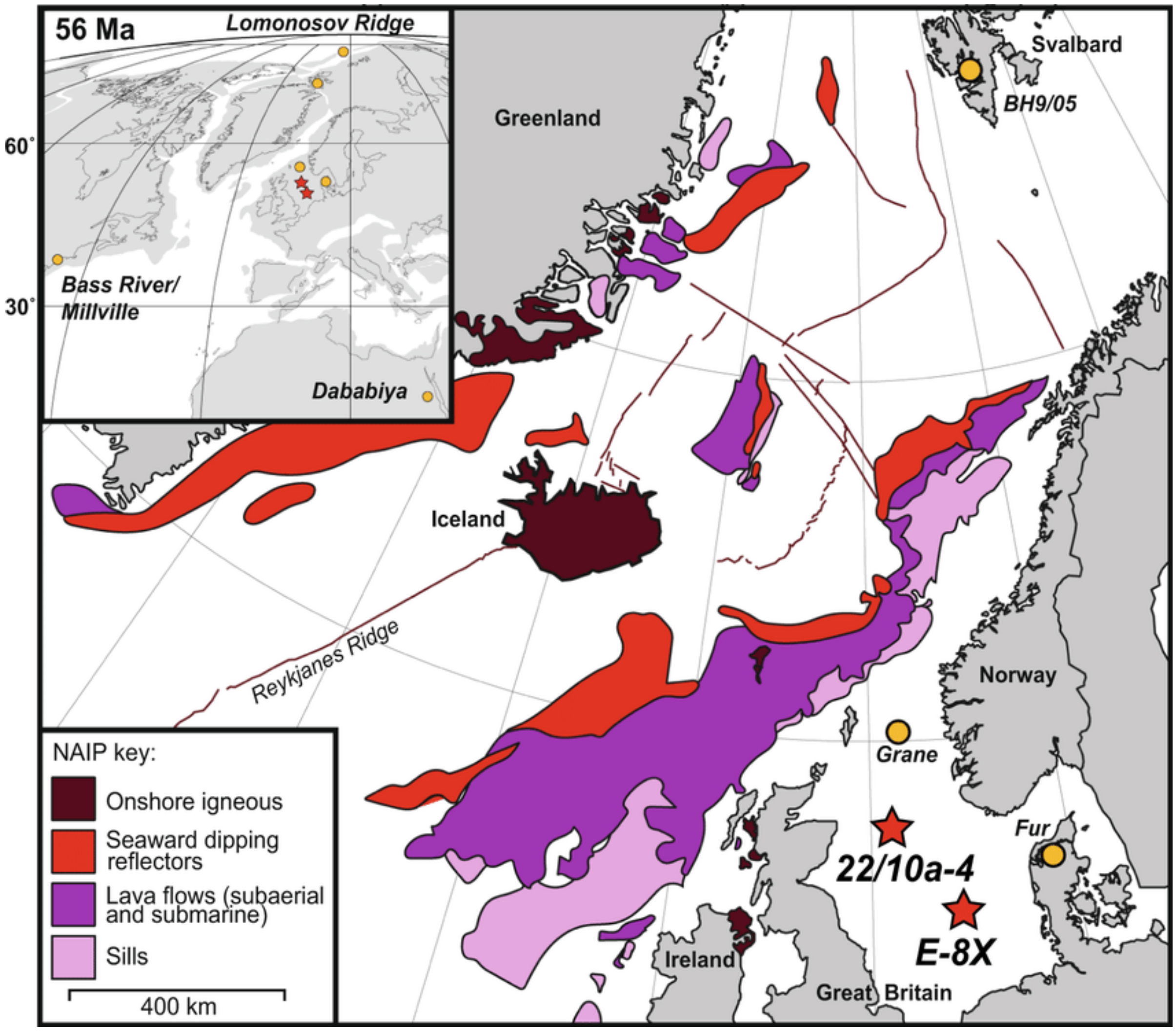

Figure 2 - This map describes where the onshore igneous rocks formed. Notice that in the North of Ireland, where Giant's Causeway is located, there is indication of onshore igneous rock.

Adaptation of the Geologic Time Scale During which the Basalt Rock Formed at Giant’s Causeway

Figure 3

Adaptation of the Geologic Time Scale, taken from the International Commission on Stratigraphy

Figure 3 - Adapted geologic time scale, based on estimations and data from Giant's Causeway when it comes to the time period

For the basalt rocks present at Giant’s Causeway, the formation was only from the magma that emerged during this epoch, where Europe and North America were splitting apart. From this magma, the igneous rock of basalt was created through the process of cooling and recrystallization, and the hexagonal columns were created through the process of columnar jointing, where lava flows would cool and would create contractions/fracturing. Additionally, basalt rock formed during the Paleogene system, within the Paleocene epoch, so the numeric age is ~56-62 million years ago, consistent with the findings that have been presented.

Conclusions about Giant's Causeway

From this research, I've concluded that Giant's Causeway was part of a major volcanism event and was part of the North Atlantic Igneous Province, which had major volcanic activity back in the Paleocene epoch. Additionally, I learnt about a different kind of radiometric dating technique utilizing argon isotopes instead of the traditional potassium-argon isotope method to accurately estimate the age of the basalt rocks present at Giant's Causeway. I also learnt that, through the law of superposition, that sediment that is older than the basalt rock means that the basalt rock must be younger than the sediment itself.

Throughout this research project, a major process that I learnt about when it came to the formation of the unique hexagonal basalt columns was of columnar jointing, where contractions and fracturing of lava flows would form these shapes. Additionally, I learnt that not all the rock cycle needs to be used to form specific rocks like basalt; a fraction of it may only be used depending on the history of formation. Overall, this research project taught me that there is much depth to geology and Earth in general than meets the eye, and how a place like Giant's Causeway in Northern Ireland can hold so much significance and history about the formation of our Earth just through the analysis of its rocks present at the site.

References

Argon-argon dating - an overview | sciencedirect topics. Science Direct. (2007). https://www.sciencedirect.com/topics/earth-and-planetary-sciences/argon-argon-dating

Earth Science Conservation Review - Giant's Causeway. ESCR site summary. (2025). https://www2.habitas.org.uk/escr/summary.php?item=533

Giant's Causeway's History. National Trust. (2025). https://www.nationaltrust.org.uk/visit/northern-ireland/giants-causeway/history-of-giants-causeway

Hull Geological Society. The Tertiary Basalts of Antrim and the Underlying Chalk – The Giant's Causeway. (2012). https://www.hullgeolsoc.co.uk/hg1506.htm

International Commission on Stratigraphy - Chart. International Commission on stratigraphy. (2025). https://stratigraphy.org/chart

Oxford Academic. (2010). *North Atlantic Igneous Province reconstructed and its relation to the Plume Generation Zone: The Antrim Lava Group Revisited | Geophysical Journal International | oxford academic*. Geophysical Journal International. https://academic.oup.com/gji/article/182/1/183/560521

Paleocene Eocene Carbon Feedbacks Triggered by Volcanic Activity. Research Gate. (2021). https://www.researchgate.net/publication/354247454_PaleoceneEocene_carbon_feedbacks_triggered_by_volcanic_activity

The paleocene volcanic rocks of the giant's causeway and Causeway Coast. IUGS. (2022, September 26). https://iugs-geoheritage.org/geoheritage_sites/the-paleocene-volcanic-rocks-of-the-giants-causeway-and-causeway-coast/

Paleocene-Eocene Thermal Maximum. Encyclopedia Britannica. (2025). https://www.britannica.com/science/Paleocene-Eocene-Thermal-Maximum